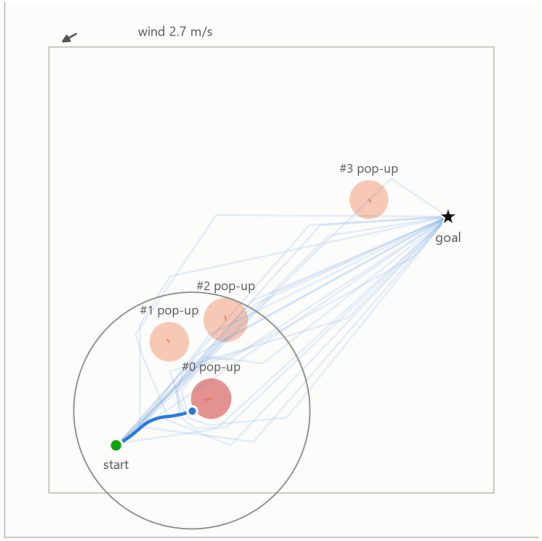
One real case, end to end

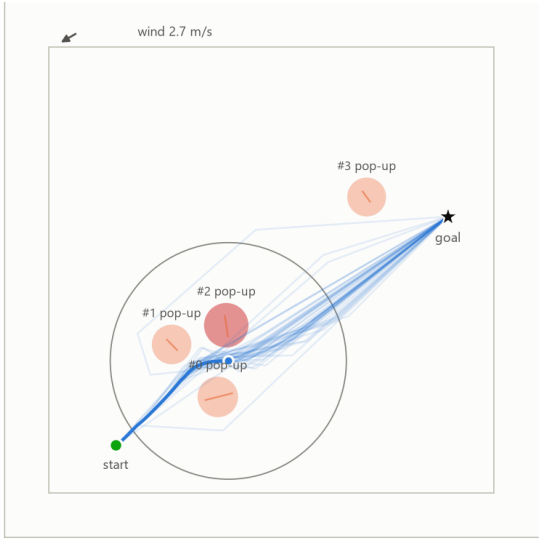
A crash generated by the diffusion red team is measured, explained, fixed, re-flown, and then the red team attacks the fixed system again. Every picture and number on these pages is produced by `experiments/case_study.py` from the project's stored runs; the LLM text is Gemini's actual answer.

1 Crash created	2 Measured causes	3 LLM explanation	4 Fix and re-fly	5 Attacked again
Diffusion + feedback (round 4) generates a scenario; RRT collision.	Six one-factor re-runs: **UAV too fast to react + unlucky planner randomness**.	Gemini names uav_speed, planner_randomness; every citation checks out.	**speed governor** switched on; the same scenario now reaches the goal.	Red team finds a new crash: **wind / gusts + late detection (short sensing range) + sensor noise**.

original system: crash



hardened system: reaches the goal



The short version.

Left: the generated scenario on the original planner stack (crash). Right: the same scenario, same RRT random seed, after the hardening derived from the measured causes. Minimum clearance -0.17 m → 1.31 m.

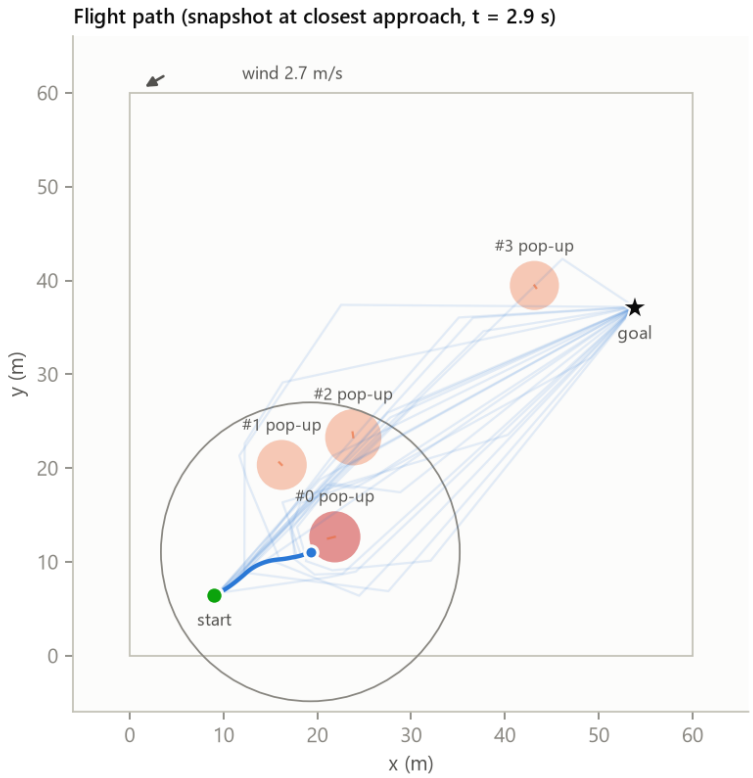
The loop does not end there: page 6 shows the next crash the red team found against the hardened system.

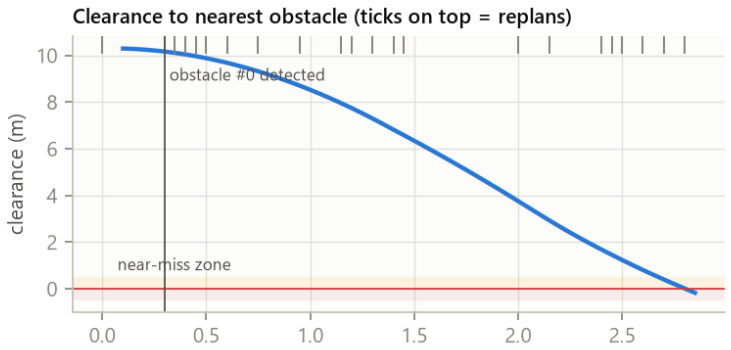
Step 1 · The red team creates a crash

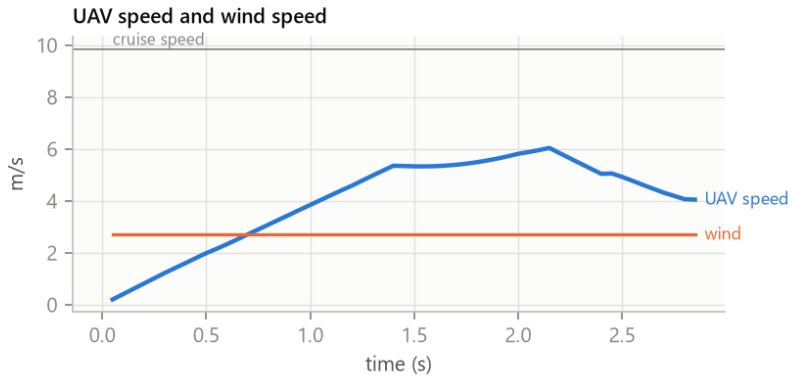
The conditional diffusion model, fine-tuned on failures it found in earlier rounds (this scenario is from round 4), generates a scenario (46 numbers: obstacles, wind, gusts, sensing). The unmodified PythonRobotics RRT flies it in the simulator. Status: **collision**, failure score F = 0.90.

Step 1: the generated scenario, original system

RRT, status: collision







cruise speed	obstacles (moving)	wind	gust	sensing range	sensor noise
9.9 m/s	4 (100%)	headwind, 2.7 m/s	none	15.9 m	0.4 m

Step 2 · Measure why it crashed

The simulator re-flies the identical scenario six times, changing one factor each time (same planner seed). A change that turns the crash into a safe flight is a **measured cause**. This is evidence, not a guess, and it is what the LLM is given.

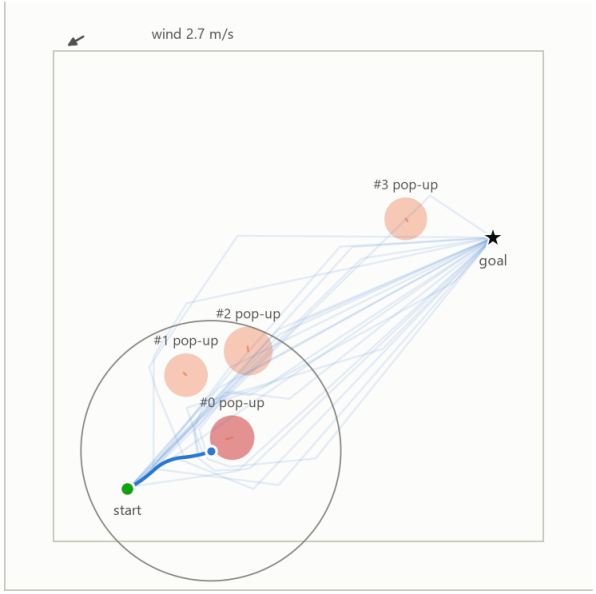
Same scenario re-flown six times, one factor changed each time

wind and gusts removed	still crashes
sensing range raised to 25 m	still crashes
sensor noise removed	still crashes
pop-up obstacles present from the start	still crashes
cruise speed reduced by 30%	**crash prevented → measured cause**
re-planned with 3 other RRT seeds	**crash prevented → measured cause**

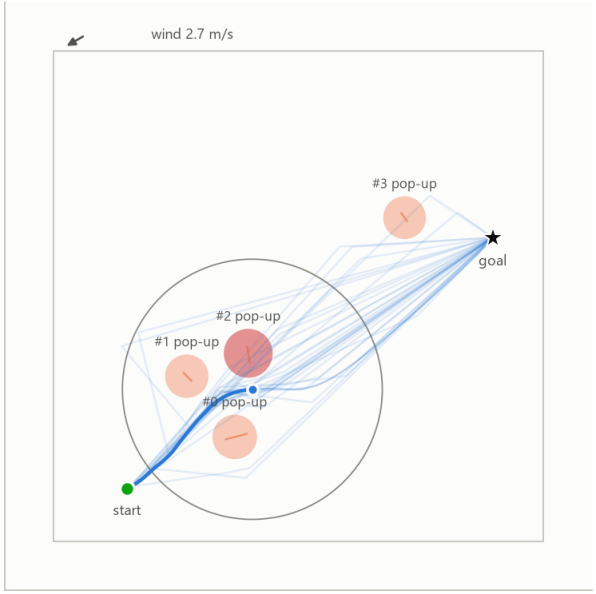
Measured failure type: UAV too fast to react + unlucky planner randomness

Only these changes turn the crash into a safe flight, so they are the measured causes. Below: the original flight next to the re-run with the first of them (cruise speed reduced by 30%).

original scenario: crash



UAV cruise speed reduced by 30%: safe



Step 3 · The LLM explains it, and a script checks the explanation

Gemini receives only this case's evidence record: the scenario, what the simulator measured, and the counterfactual results. It must name factors from a fixed vocabulary and cite, for each, the field of the record that supports it. A script then verifies every citation.

Evidence record (excerpt)

field	value
outcome / end status	failure / collision
min clearance	-0.17 m at t = 2.9 s
closest obstacle detected at	0.3 s (gap 10.6 m)
detection → closest approach	2.5 s
max wind felt	2.7 m/s
replans / planner failures	21 / 0
counterfactuals.no_wind	still fails
counterfactuals.full_sensing	still fails
counterfactuals.no_noise	still fails
counterfactuals.no_popups	still fails
counterfactuals.slower	failure prevented
counterfactuals.reseed	failure prevented

gemini-3.1-flash-lite · failure type: **collision** · confidence: high

"The UAV collided with a moving obstacle at 2.9 seconds, resulting in a minimum clearance of -0.17 meters. The failure was primarily driven by the UAV's cruise speed, as reducing it by 30% prevented the collision. Additionally, the flight was sensitive to planner randomness, as re-seeding the RRT algorithm also prevented the failure."

Factors and the evidence they cite

- **uav_speed** cites `counterfactuals.slower`
- **planner_randomness** cites `counterfactuals.reseed`

Grounding check (script): PASSED: every cited field exists in this case's evidence record; outcome copied correctly

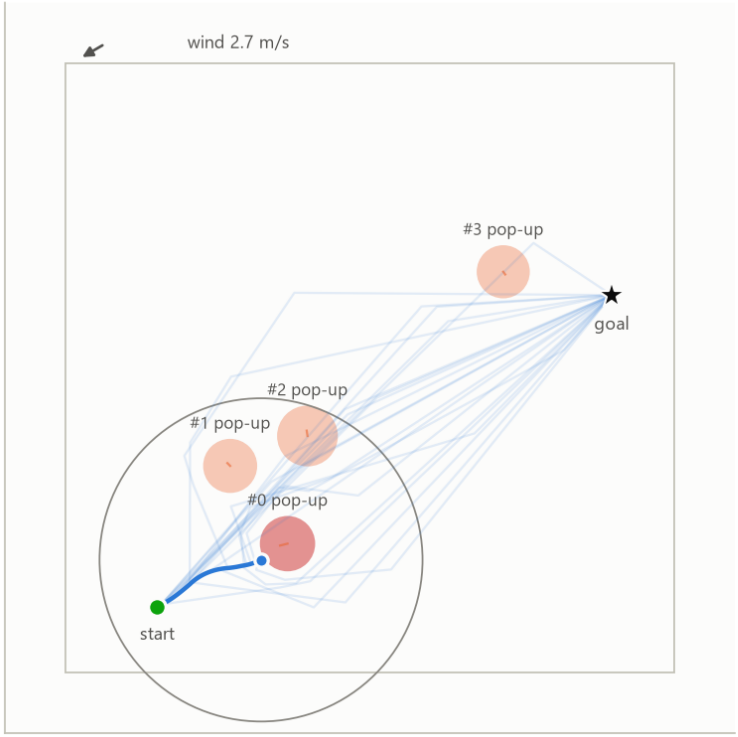
Why the counterfactuals matter. Across 50 scored crashes, the same LLM given only the flight log names 34% of the measured causes and contradicts them in 44% of answers; given the counterfactual results as well, it names 96% with 2% contradictions (report, Table 3).

Step 4 · Fix the cause, fly the same scenario again

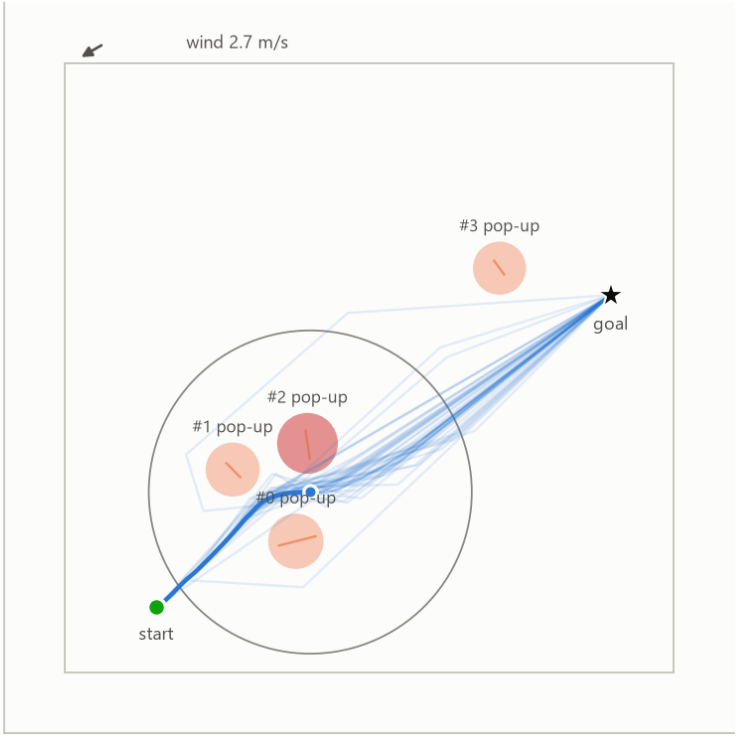
Hardening changes only the code around RRT (controller, speed limit, number of planning attempts, obstacle sweep); the RRT algorithm itself is untouched. Each option targets a cause measured across all failures. The one that targets this case's cause is highlighted: **speed governor**, speed is capped so the UAV can stop within its sensing range.

hardening option	targets measured cause	what it does
wind feed-forward	wind / gusts	controller cancels the estimated wind force
speed governor	UAV too fast / late detection	speed capped so the UAV can stop within sensing range
3 planning attempts	unlucky planner randomness	RRT run 3x per replan, shortest path kept
longer obstacle sweep	moving / pop-up obstacles	moving obstacles swept 2 s ahead (was 1 s)

original system: crash



hardened system: reaches the goal

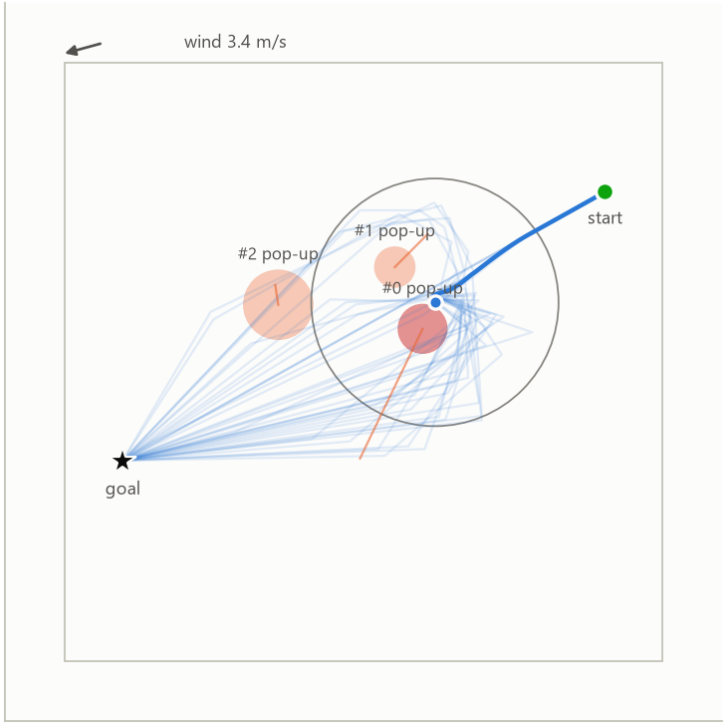


Same scenario, same RRT seed. Original: **collision**, min clearance -0.17 m. Hardened: **goal**, min clearance 1.31 m, flight time 13.6 s. Across all 8,650 failures found so far, the hardened system survives 79%.

Step 5 · The red team attacks the hardened system

Diffusion + feedback is run again, now against the hardened system. Failures are rarer (the failure rate on random scenarios fell from 12.0% to 4.7%), but the red team still finds them, 2.1× faster than random sampling. Here is one, analysed exactly like the first.

hardened system: new crash



New crash, re-flown on the hardened system with one factor changed each time

wind and gusts removed	**crash prevented → measured cause**
sensing range raised to 25 m	**crash prevented → measured cause**
sensor noise removed	**crash prevented → measured cause**
pop-up obstacles present from the start	still crashes
cruise speed reduced by 30%	still crashes
re-planned with 3 other RRT seeds	still crashes

gemini-3.1-flash-lite · failure type: **collision** · confidence: high

> *"The UAV collided with a moving obstacle at 5.8 seconds due to a combination of environmental and sensing limitations. The flight failure was confirmed to be prevented by removing the 3.4 m/s tailwind, increasing the sensing range beyond 12.4 m, or eliminating the 0.32 m sensor noise."*

Factors and the evidence they cite

- **wind** cites `counterfactuals.no_wind`
- **late_detection** cites `counterfactuals.full_sensing`
- **sensor_noise** cites `counterfactuals.no_noise`

Grounding check (script): PASSED: every cited field exists in this case's evidence record; outcome copied correctly

What happens next: even with all four fixes switched on, removing any one of these prevents this crash: wind / gusts, late detection (short sensing range), sensor noise. The existing fixes do not cover this combination, so it becomes the target of the next round of hardening, and the loop continues.